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Literature Review

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2026/02/27

Article

Optical control over topological Chern number in moiré materials

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Check for updates

O. Huber[1], K. Kuhlbrodt[1], E. Anderson[2], W. Li[2], K. Watanabe[3], T. Taniguchi[4], M. Kroner[1], X. Xu[2], A. Imamoğlu[1]✉ & T. Smoleński[5]✉

Nature 649, pages 1153–1158 (2026)

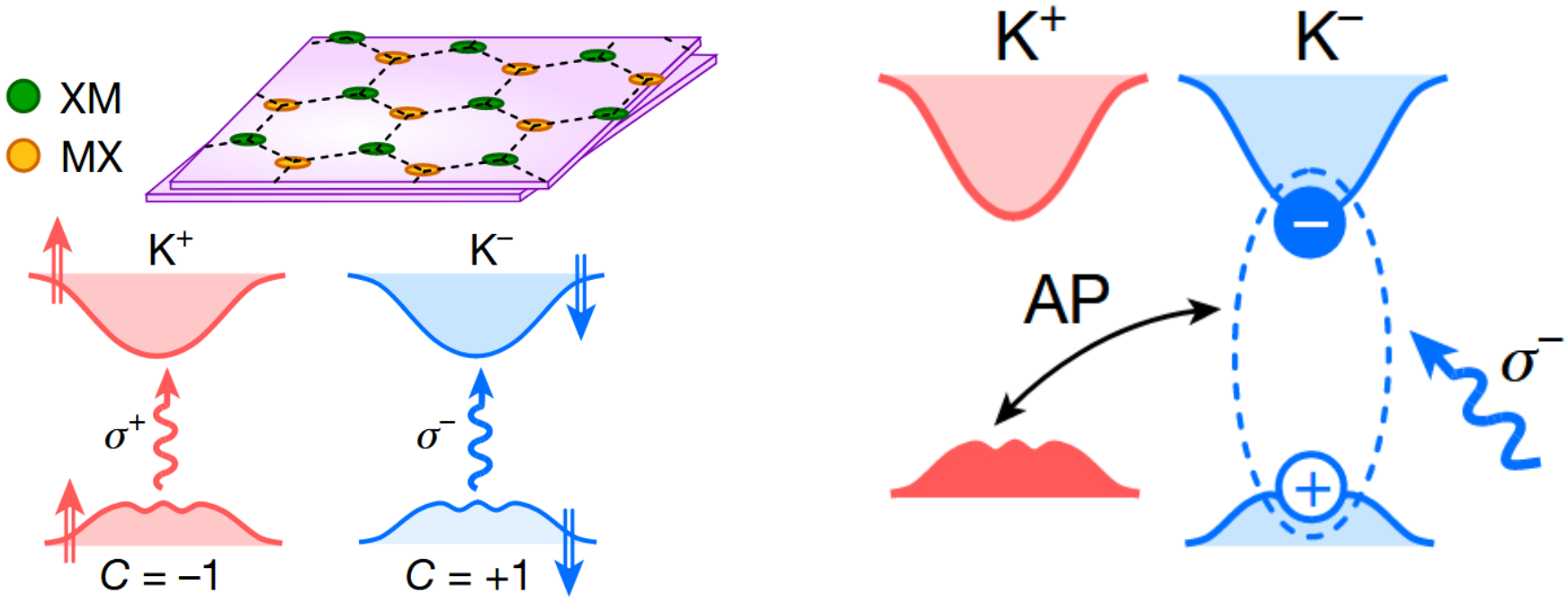
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01.

Background

Attractive polaron as a spin sensor

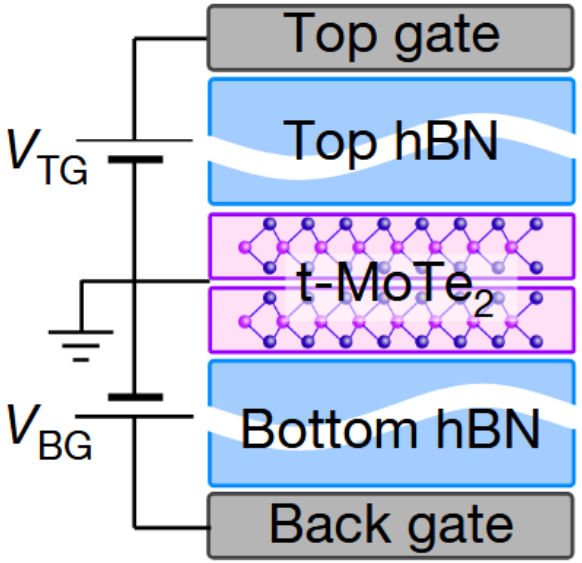
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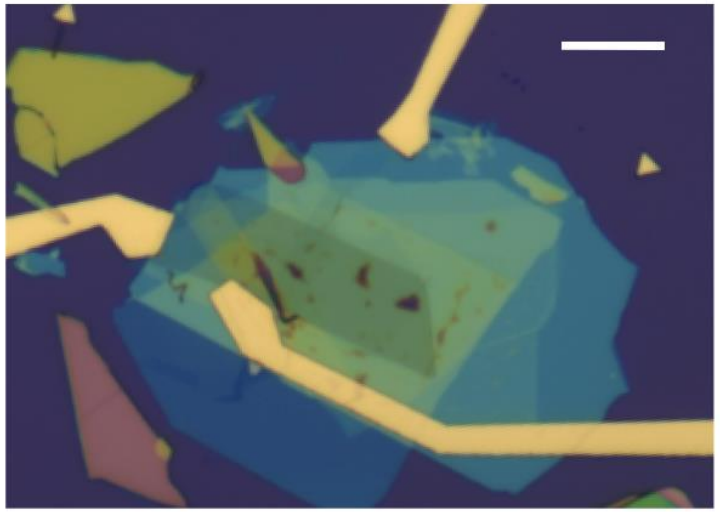


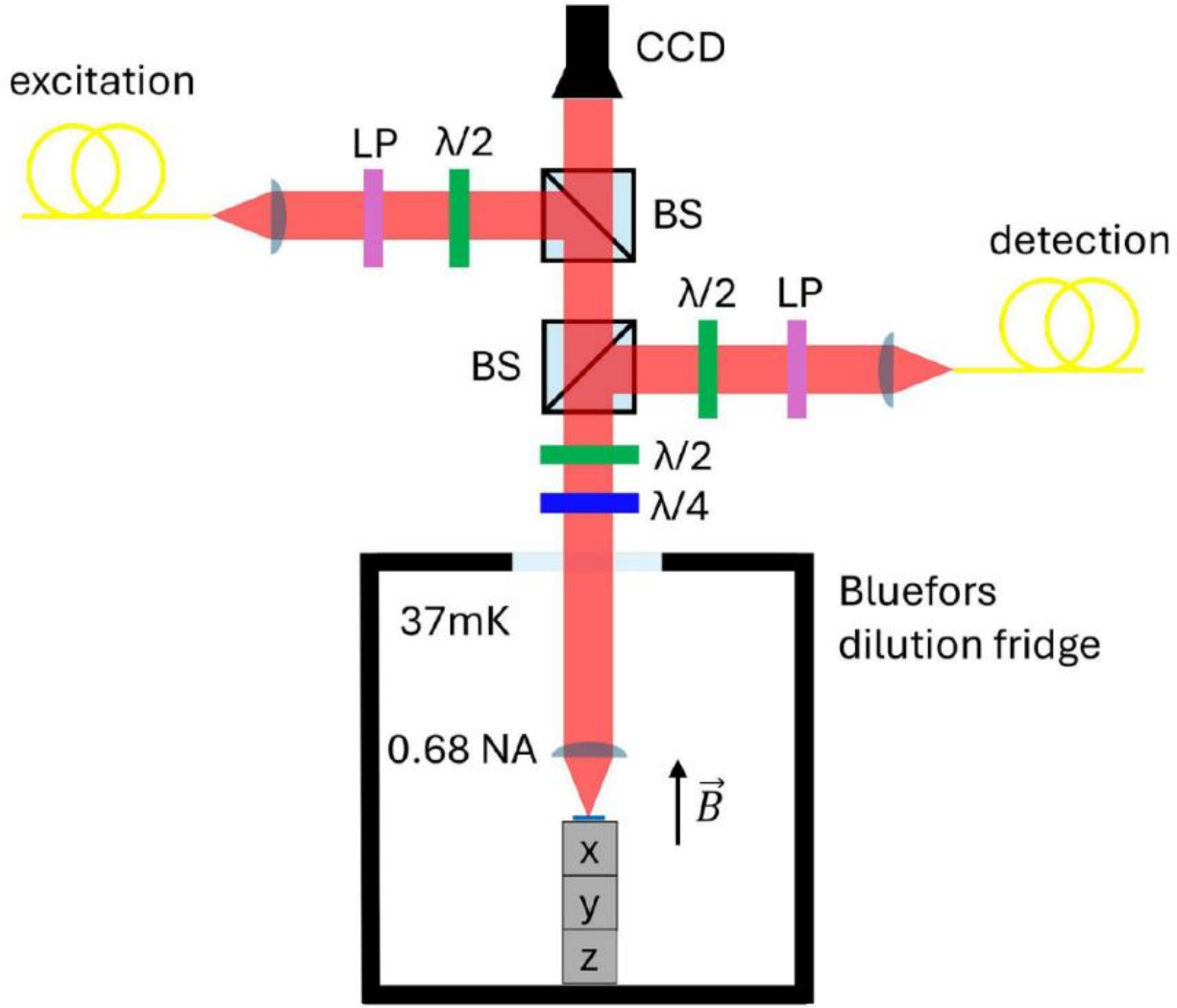
Ferromagnetism + Spin-valley selection = “Valley ferromagnetism”

Device and set-up

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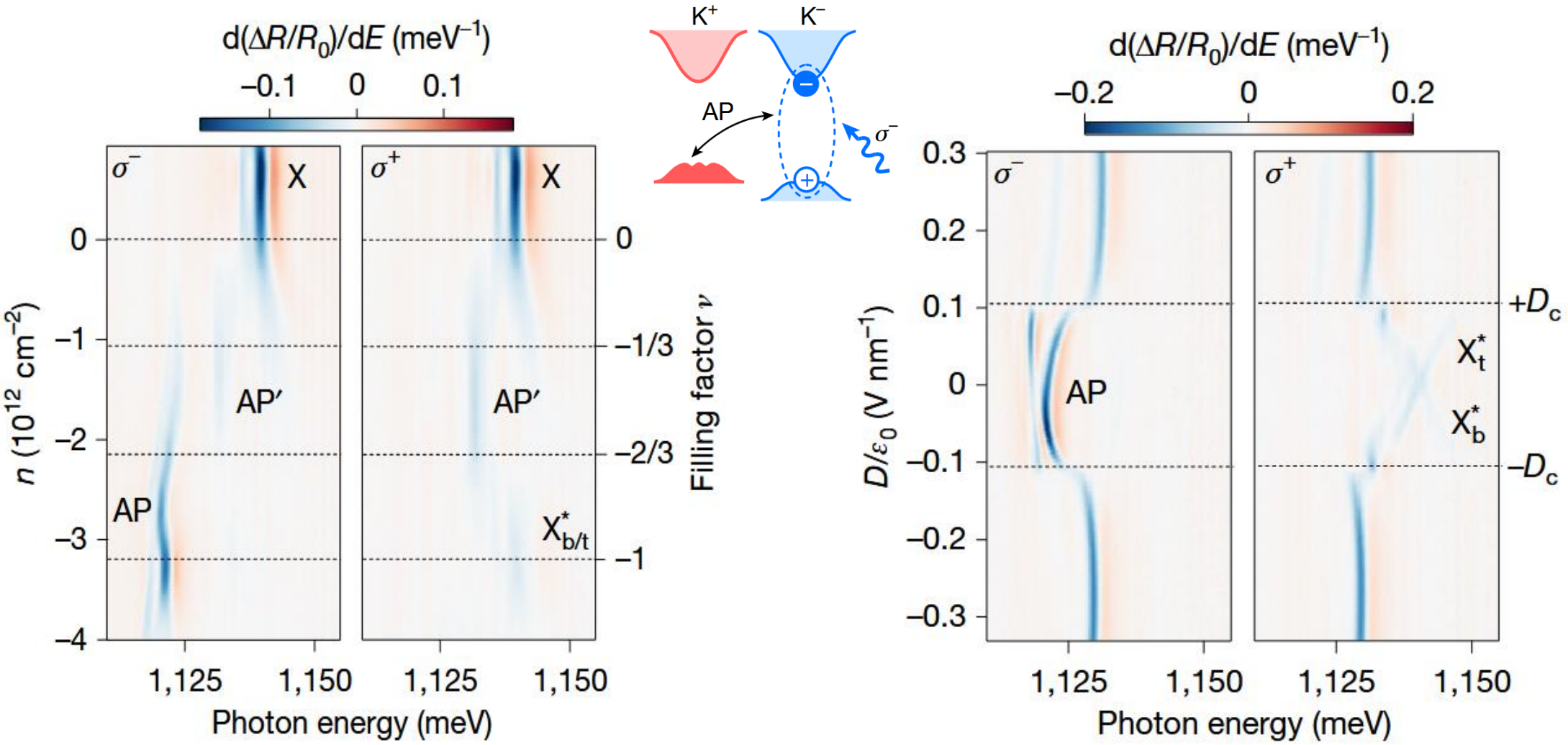
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02.

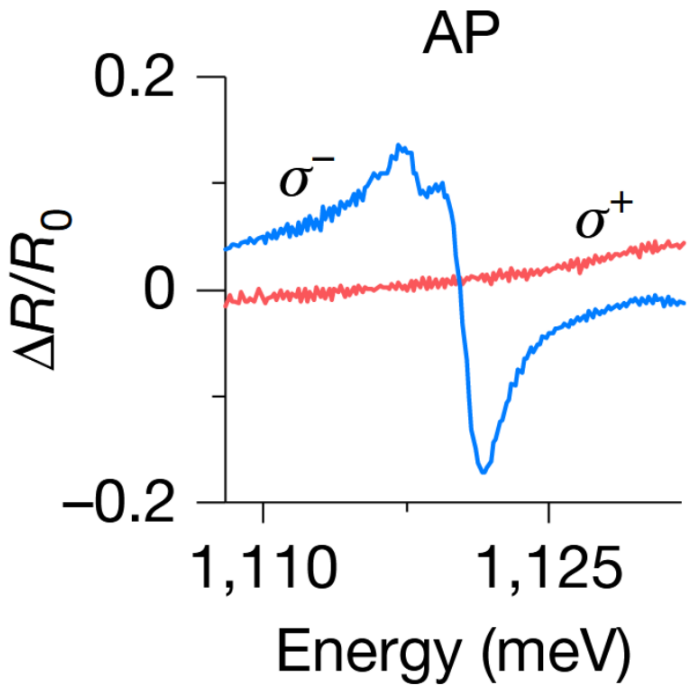
Measurement

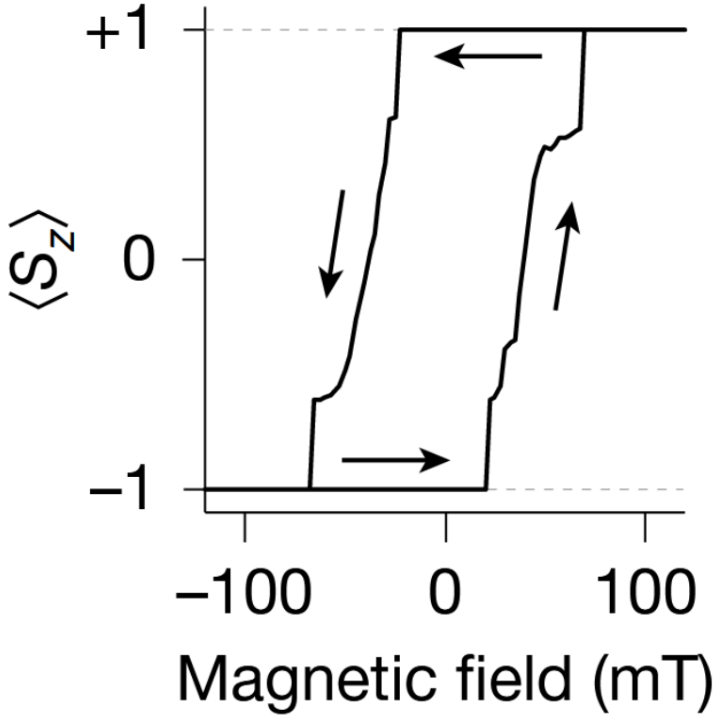
Characterization

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Ferromagnetic



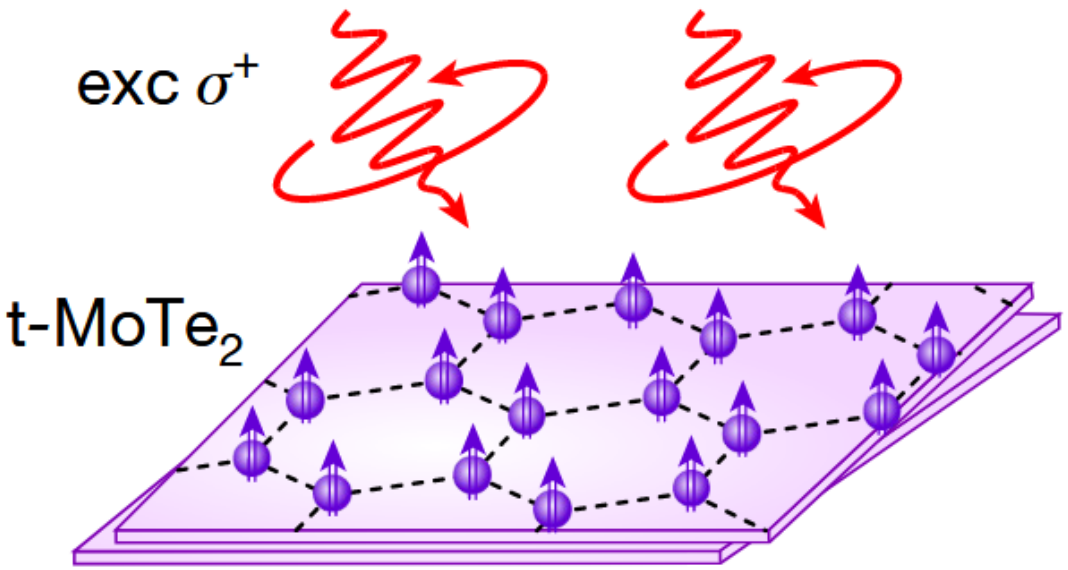


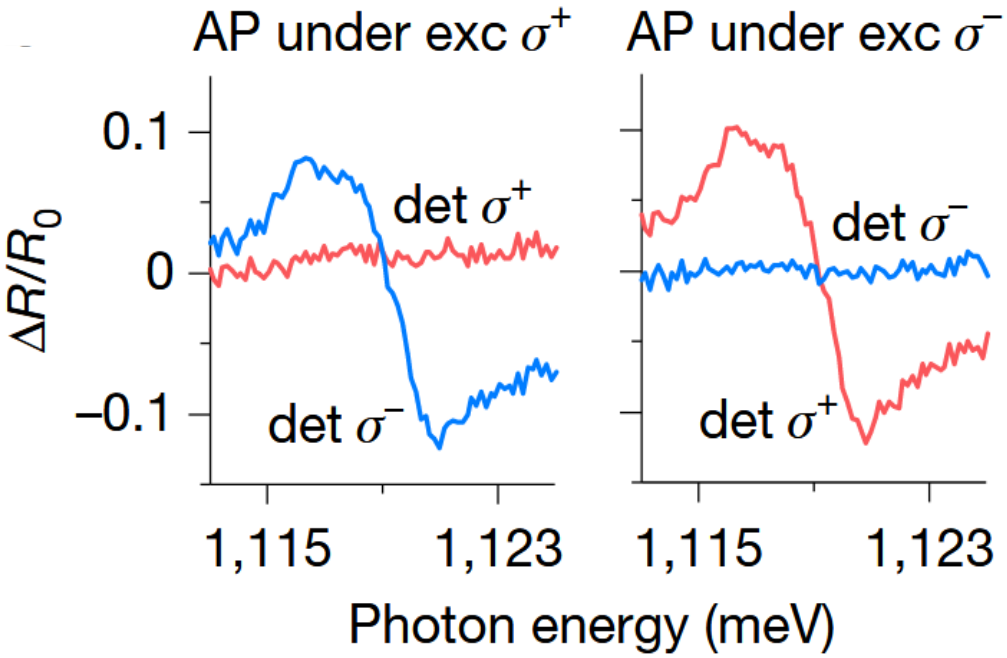
Degree of AP circular polarization $\rho_{\mathrm{AP}} = (f_{\mathrm{AP}}^{-} - f_{\mathrm{AP}}^{+})/(f_{\mathrm{AP}}^{-} + f_{\mathrm{AP}}^{+})$

Degree of spin–valley polarization $\langle S_z \rangle = (n_+ - n_-)/(n_+ + n_-)$

Optical control over Chern number

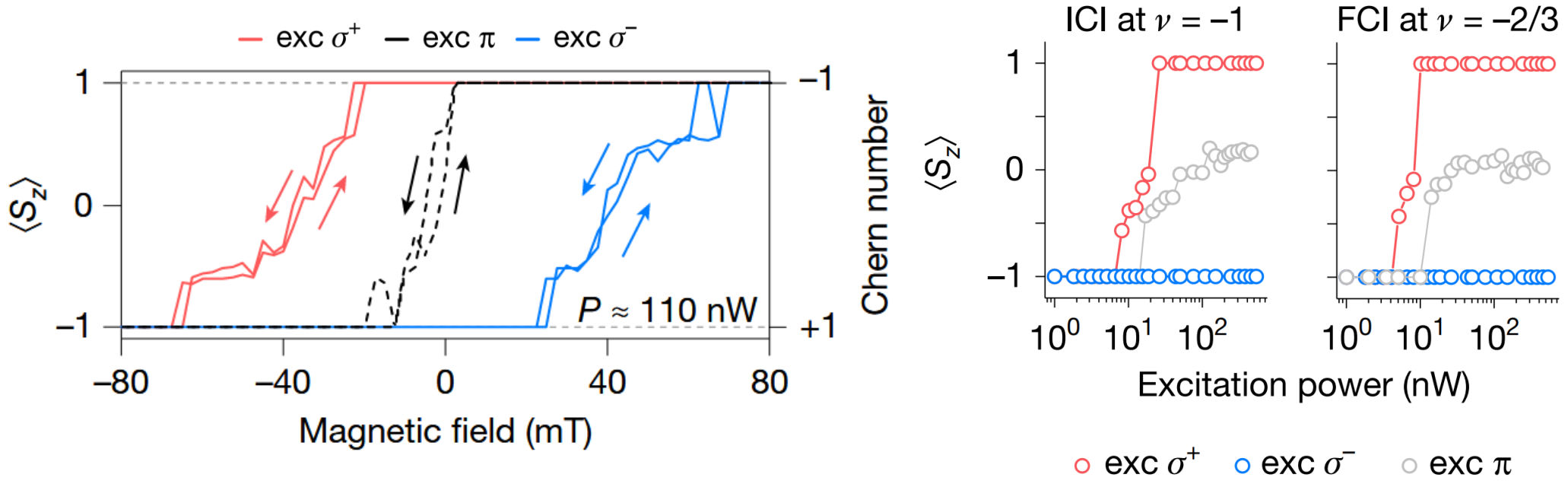
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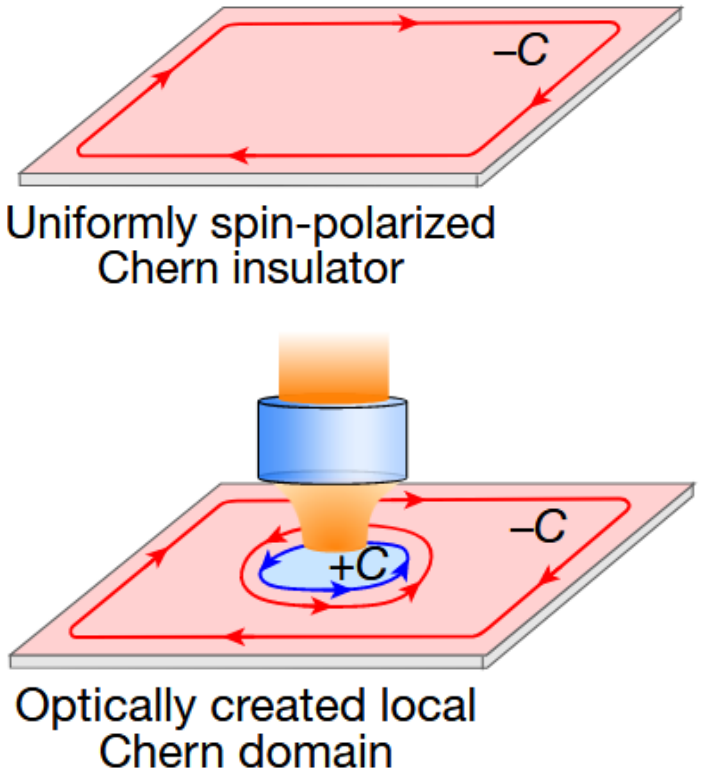
Optical control over Chern number

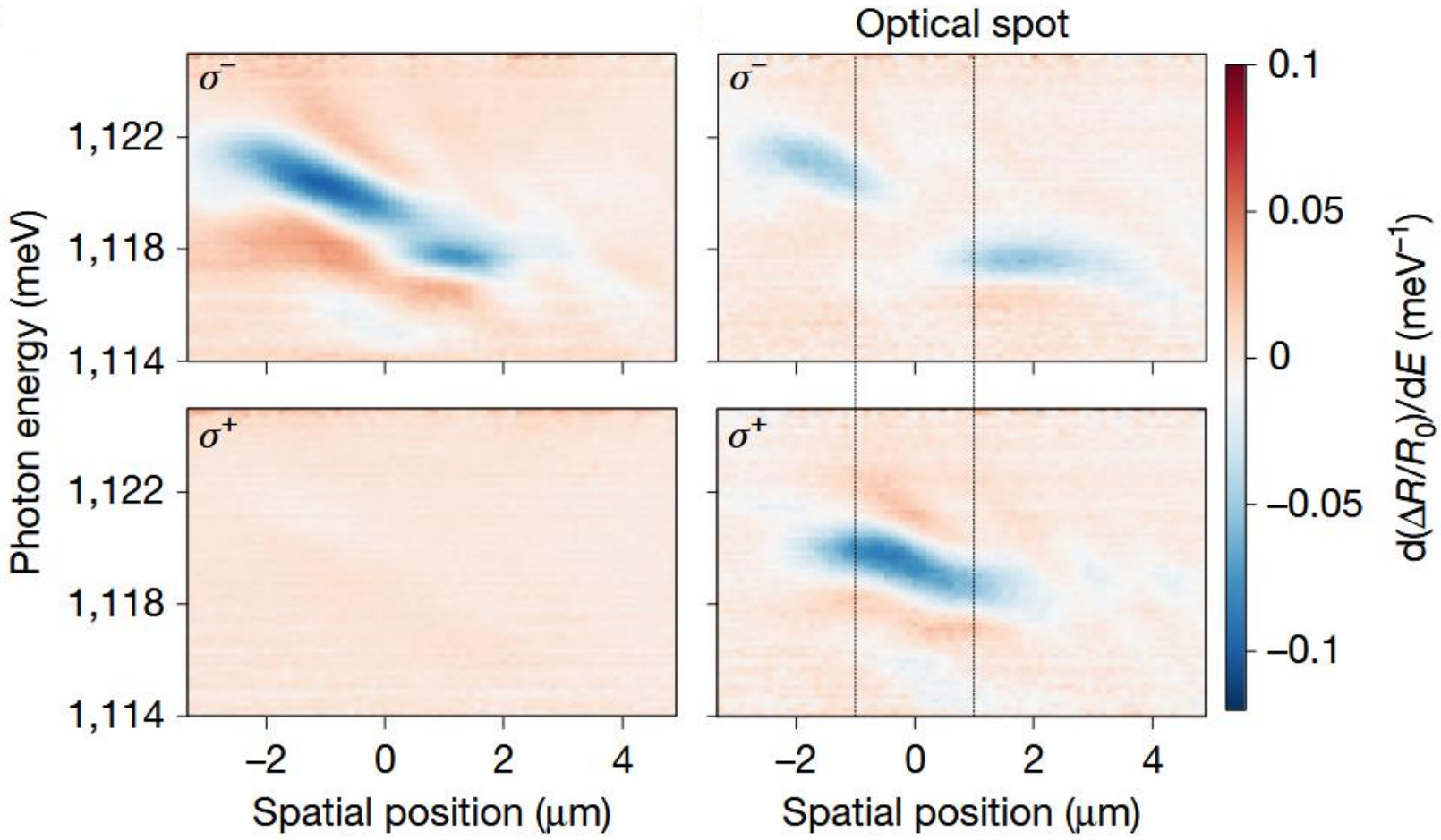
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Localized optical writing

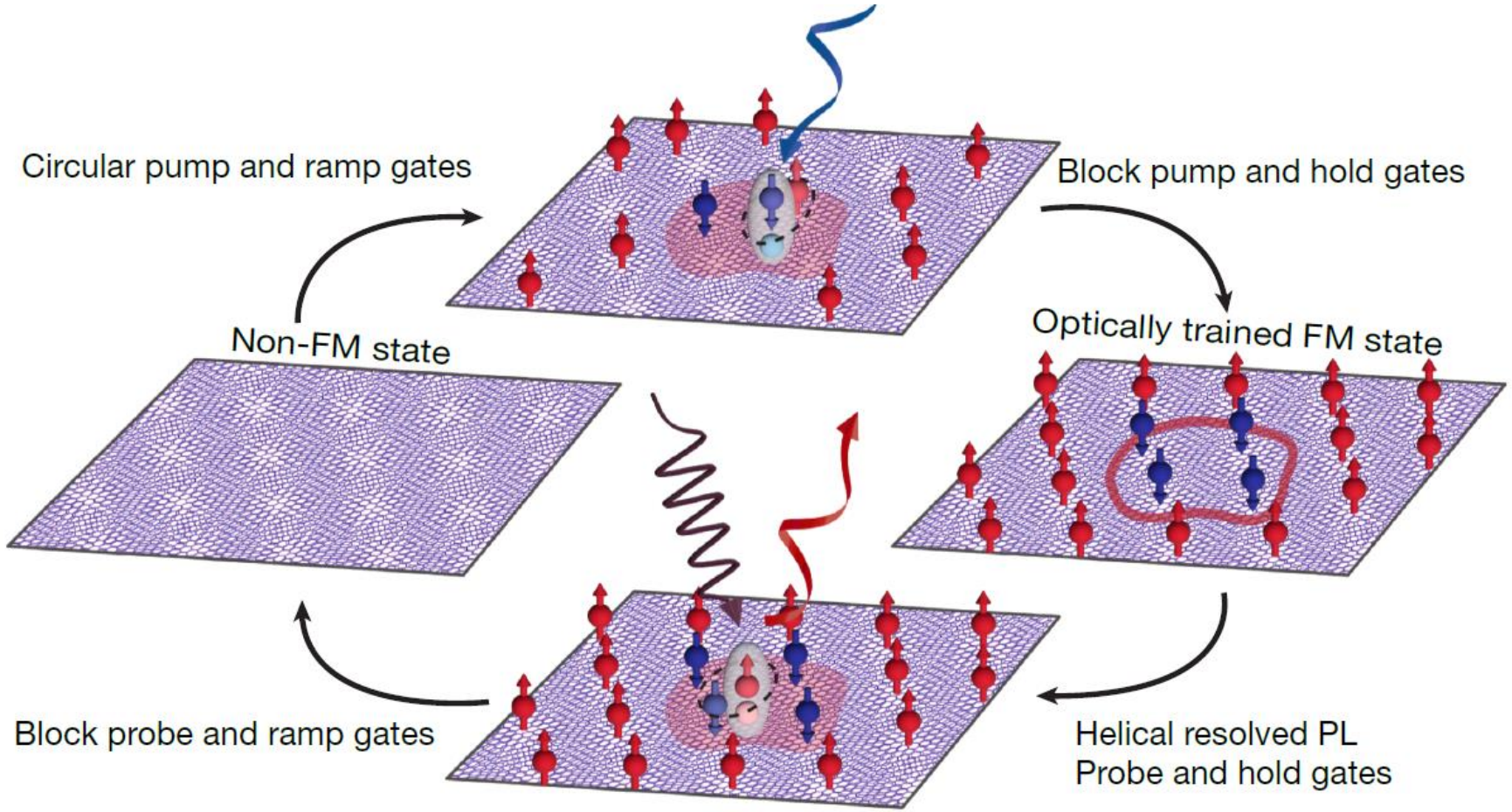
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THANKS

Speaker: Zhongqi Xiu